

Reto 3 Enunciado:

Implementa una base de datos en ORACLE sobre un tema que te interese. Debe cumplir:

- Es necesario que implemente relaciones 1:1, 1:N y M:N.
- Se deben usar todos los tipos de datos vistos en clase.
- Se insertarán al menos 3 registros en cada tabla mediante SQL.

1.- Realizar el sistema de tablas y la inserción de datos. (2 PUNTOS)

- PEGA AQUÍ LAS SENTENCIAS DDL Y DML –
SENTENCIAS DDL

TABLA CHEF

```
CREATE TABLE CHEF(  
  ID NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
  NOMBRE VARCHAR2(40) NOT NULL,  
  EMAIL VARCHAR2(80) UNIQUE,  
  TELEFONO VARCHAR2(15),  
  FECHA_ALTA DATE DEFAULT SYSDATE  
);
```

TABLA CHEF_PERFIL 1:1 con CHEF

```
CREATE TABLE CHEF_PERFIL(  
  ID NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
  ID_CHEF NUMBER NOT NULL UNIQUE,  
  FECHA_NACIMIENTO DATE,  
  NACIONALIDAD VARCHAR2(30),  
  EXPERIENCIA_ANIOS NUMBER(2) CHECK (EXPERIENCIA_ANIOS BETWEEN 0 AND 60),  
  FOREIGN KEY (ID_CHEF) REFERENCES CHEF(ID)  
);
```

TABLA RECETA

```
CREATE TABLE RECETA(  
  ID NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
  TITULO VARCHAR2(60) NOT NULL,  
  FECHA_PUBLICACION DATE,  
  RATING FLOAT,  
  ID_CHEF NUMBER NOT NULL,  
  FOREIGN KEY (ID_CHEF) REFERENCES CHEF(ID)  
);
```

TABLA PASO 1:N con RECETA

```
CREATE TABLE PASO(  
  ID NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
  NUM_PASO NUMBER(3) NOT NULL,  
  DESCRIPCION VARCHAR2(200) NOT NULL,  
  DURACION_MIN NUMBER(3) CHECK (DURACION_MIN BETWEEN 0 AND 999),  
  ID_RECETA NUMBER NOT NULL,  
  FOREIGN KEY (ID_RECETA) REFERENCES RECETA(ID)  
);
```

TABLA INGREDIENTE

```
CREATE TABLE INGREDIENTE(  
  ID NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
  NOMBRE VARCHAR2(40) NOT NULL,  
  UNIDAD CHAR(2) NOT NULL,  
  PRECIO_MEDIO FLOAT  
);
```

TABLA RECETA INGREDIENTE (M:N) INTERMEDIA ENTRE RECETA E INGREDIENTE

```
CREATE TABLE RECETA_INGREDIENTE(  
  ID NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
  ID_RECETA NUMBER NOT NULL,  
  ID_INGREDIENTE NUMBER NOT NULL,  
  CANTIDAD NUMBER(7,2) NOT NULL,  
  FOREIGN KEY (ID_RECETA) REFERENCES RECETA(ID),  
  FOREIGN KEY (ID_INGREDIENTE) REFERENCES INGREDIENTE(ID)  
);
```

SENTENCIAS DML

CHEF (3)

```
INSERT INTO CHEF (NOMBRE, EMAIL, TELEFONO, FECHA_ALTA)  
VALUES ('Lucia Ramos', 'lucia.ramos@uni.com', '+34-600111222', TO_DATE('2025-01-10','YYYY-MM-DD'));  
  
INSERT INTO CHEF (NOMBRE, EMAIL, TELEFONO, FECHA_ALTA)  
VALUES ('Mario Vega', 'mario.vega@uni.com', '+34-600333444', TO_DATE('2025-02-03','YYYY-MM-DD'));  
  
INSERT INTO CHEF (NOMBRE, EMAIL, TELEFONO, FECHA_ALTA)  
VALUES ('Sara Nunez', 'sara.nunez@uni.com', '+34-600555666', TO_DATE('2025-03-15','YYYY-MM-DD'));
```

CHEF_PERFIL aquí el ID_CHEF es FK y UNIQUE para mantener la relación 1:1

```
INSERT INTO CHEF_PERFIL (ID_CHEF, FECHA_NACIMIENTO, NACIONALIDAD,
EXPERIENCIA_ANIOS)
VALUES (1, TO_DATE('1998-04-21','YYYY-MM-DD'), 'Espana', 4);

INSERT INTO CHEF_PERFIL (ID_CHEF, FECHA_NACIMIENTO, NACIONALIDAD,
EXPERIENCIA_ANIOS)
VALUES (2, TO_DATE('1995-11-02','YYYY-MM-DD'), 'Italia', 6);

INSERT INTO CHEF_PERFIL (ID_CHEF, FECHA_NACIMIENTO, NACIONALIDAD,
EXPERIENCIA_ANIOS)
VALUES (3, TO_DATE('2000-07-30','YYYY-MM-DD'), 'Mexico', 2);
```

RECETA

```
INSERT INTO RECETA (TITULO, FECHA_PUBLICACION, RATING, ID_CHEF)
VALUES ('Pasta carbonara', TO_DATE('2025-04-01','YYYY-MM-DD'), 4.6, 2);

INSERT INTO RECETA (TITULO, FECHA_PUBLICACION, RATING, ID_CHEF)
VALUES ('Tortilla de patatas', TO_DATE('2025-04-10','YYYY-MM-DD'), 4.3, 1);

INSERT INTO RECETA (TITULO, FECHA_PUBLICACION, RATING, ID_CHEF)
VALUES ('Tacos de pollo', TO_DATE('2025-04-20','YYYY-MM-DD'), 4.8, 3);
```

PASO

```
INSERT INTO PASO (NUM_PASO, DESCRIPCION, DURACION_MIN, ID_RECETA)
VALUES (1, 'Preparar ingredientes y utensilios.', 5, 1);

INSERT INTO PASO (NUM_PASO, DESCRIPCION, DURACION_MIN, ID_RECETA)
VALUES (1, 'Pelar y cortar patatas, calentar aceite.', 10, 2);

INSERT INTO PASO (NUM_PASO, DESCRIPCION, DURACION_MIN, ID_RECETA)
VALUES (1, 'Cortar pollo y preparar especias.', 7, 3);
```

INGREDIENTE

```
INSERT INTO INGREDIENTE (NOMBRE, UNIDAD, PRECIO_MEDIO)
VALUES ('Huevo', 'ud', 0.30);

INSERT INTO INGREDIENTE (NOMBRE, UNIDAD, PRECIO_MEDIO)
VALUES ('Pasta', 'g', 1.20);

INSERT INTO INGREDIENTE (NOMBRE, UNIDAD, PRECIO_MEDIO)
VALUES ('Pollo', 'g', 4.50);
```

RECETA_INGREDIENTE

```
INSERT INTO RECETA_INGREDIENTE (ID_RECETA, ID_INGREDIENTE, CANTIDAD)
VALUES (1, 2, 200);
```

```
INSERT INTO RECETA_INGREDIENTE (ID_RECETA, ID_INGREDIENTE, CANTIDAD)
VALUES (2, 1, 4);
```

```
INSERT INTO RECETA_INGREDIENTE (ID_RECETA, ID_INGREDIENTE, CANTIDAD)
VALUES (3, 3, 180);
```

2.- Generar las vistas que unan todas las tablas. (3 PUNTOS)

PEGAR EL CÓDIGO SQL DE CREACIÓN DE CADA VISTA. SE PIDE UNA VISTA CON LA RELACIÓN 1:N, OTRA CON LA RELACIÓN M:N Y OTRA CON TODO EL SISTEMA COMPLETO. USA INNER JOIN.

- PEGA AQUÍ LAS SENTENCIAS SQL -

VISTA RELACIÓN 1:N RECETA – PASOS

```
CREATE VIEW V_RECETA_PASOS AS
SELECT R.ID_CHEF,
       R.TITULO,
       PA.NUM_PASO,
       PA.DESCRIPCION,
       PA.DURACION_MIN
FROM RECETA R INNER JOIN PASO PA
ON R.ID = PA.ID_RECETA;
```

```
SELECT * FROM V_RECETA_PASOS;
```

VISTA 2 RELACION M:N RECETA INGREDIENTE tabla intermedia RECETA_INGREDIENTE

```
CREATE VIEW V_RECETA_INGREDIENTES AS
SELECT R.TITULO, I.NOMBRE AS INGREDIENTE, RI.CANTIDAD, I.UNIDAD
FROM RECETA R INNER JOIN RECETA_INGREDIENTE RI
ON R.ID = RI.ID_RECETA
INNER JOIN INGREDIENTE I
ON I.ID = RI.ID_INGREDIENTE;
```

```
SELECT * FROM V_RECETA_INGREDIENTES;
```

VISTA DE TODAS LAS TABLAS

CHEF – CHEF_PERFIL(1:1),
CHEF-RECETA(1:N), RECETA-PASO (1:N),
RECETA-RECETA_INGREDIENTE -INGREDIENTE-RECETA_INGREDIENTE(M:N)

```
CREATE VIEW V_TODAS_LAS_TABLAS AS
SELECT C.ID AS ID_CHEF, C.NOMBRE AS CHEF, C.EMAIL, CP.NACIONALIDAD,
       R.TITULO AS RECETA, R.FECHA_PUBLICACION, R.RATING,
       PA.NUM_PASO, PA.DESCRIPCION,
       I.NOMBRE AS INGREDIENTE, RI.CANTIDAD, I.UNIDAD
FROM CHEF C INNER JOIN CHEF_PERFIL CP
ON C.ID = CP.ID_CHEF
INNER JOIN RECETA R
ON C.ID = R.ID_CHEF
INNER JOIN PASO PA
ON R.ID = PA.ID_RECETA
INNER JOIN RECETA_INGREDIENTE RI
ON R.ID = RI.ID_RECETA
INNER JOIN INGREDIENTE I
ON I.ID = RI.ID_INGREDIENTE;
```

```
SELECT * FROM V_TODAS_LAS_TABLAS;
```

- PEGA AQUÍ LAS CAPTURA DE PANTALLA COMPLETAS QUE MUESTREN SU CORRECTA EJECUCIÓN -

VISTA RELACIÓN 1:N RECETA PASOS

The screenshot displays the FreeSQL web application interface. On the left, a 'Navigator' pane shows a tree view of database objects under 'My Schema', including tables like ACCESOS, ACTORES, AUTORES, CAMISETAS, CHEF, CHEF_PERFIL, CLIENTES, ESTUDIANTES, EVENTOS, INFORME, INGREDIENTE, LIBROS_AUTOR, and MATERIAS. The main area contains an SQL editor with the following query:

```
1 CREATE VIEW V_RECETA_PASOS AS
2 SELECT R.ID_CHEF, R.TITULO, PA.NUM_PASO, PA.DESCRIPCION, PA.DURACION_MIN
3 FROM RECETA R INNER JOIN PASO PA
4 ON R.ID = PA.ID_RECETA;
```

Below the editor, the 'Script output' tab shows the execution results: 'Elapsed: 00:00:00.002' and 'SQL> CREATE VIEW V_RECETA_PASOS AS SELECT R.ID_CHEF, R.TITULO, PA.NUM_PASO, PA.DESCRIPCION, PA.DURACION_MIN FROM RECETA R INNER JOIN PASO PA ON R.ID = PA.ID_RECETA View V_RECETA_PASOS created. Elapsed: 00:00:00.013'. The right sidebar features a 'Library' section with search results for 'Creating Tables...' and 'Joining Tables: Databases...'. The bottom status bar indicates the system time as 18:21 on 26/01/2026.

The screenshot shows the FreeSQL web interface. In the left sidebar, the 'V_RECETA_PASOS' table is selected under the 'My Schema' section. The main area displays the 'Query result' for a SELECT query. The table has the following data:

ID_CHEF	TITULO	NUM_PASO	DESCRIPCION	DURACION_MIN
1	Pasta carbonara	1	Preparar ingrediente:	5
2	Tortilla de patatas	1	Pelar y cortar patatas:	10
3	Tacos de pollo	1	Cortar pollo y prepar.	7

VISTA RELACION M:N RECETA -INGREDIENTE T.INTERMEDIA RECETA-INCREDIENTE

The screenshot shows the FreeSQL web interface with a new SQL script being executed. The script creates a view named V_RECETA_INGREDIENTES. The 'Script output' shows the successful execution of the CREATE VIEW statement.

```

1 CREATE VIEW V_RECETA_INGREDIENTES AS
2 SELECT R.TITULO, I.NOMBRE AS INGREDIENTE, RI.CANTIDAD, I.UNIDAD
3 FROM RECETA R INNER JOIN RECETA_INGREDIENTE RI
4 ON R.ID = RI.ID_RECETA
5 INNER JOIN INGREDIENTE I
6 ON I.ID = RI.ID_INGREDIENTE;

```

The output shows:

```

SQL> CREATE VIEW V_RECETA_INGREDIENTES AS
SELECT R.TITULO, I.NOMBRE AS INGREDIENTE, RI.CANTIDAD, I.UNIDAD
FROM RECETA R INNER JOIN RECETA_INGREDIENTE RI
ON R.ID = RI.ID_RECETA...
View V_RECETA_INGREDIENTES created.
Elapsed: 00:00:00.011

```

The screenshot shows the Oracle FreeSQL web interface. The main window displays a query result table with the following data:

	TITULO	INGREDIENTE	CANTIDAD	UNIDAD
1	Pasta carbonara	Pasta	200	g
2	Tortilla de patatas	Huevo	4	ud
3	Tacos de pollo	Pollo	180	g

The interface includes a Navigator on the left with a search bar and a list of objects: NOTAS_ESTUDIANTES, V_RECETA_INGREDIENTES, and V_RECETA_PASOS. The bottom of the page shows a footer with various links and a copyright notice for 2015, 2020 Oracle and/or its affiliates.

VISTA TODAS LAS TABLAS

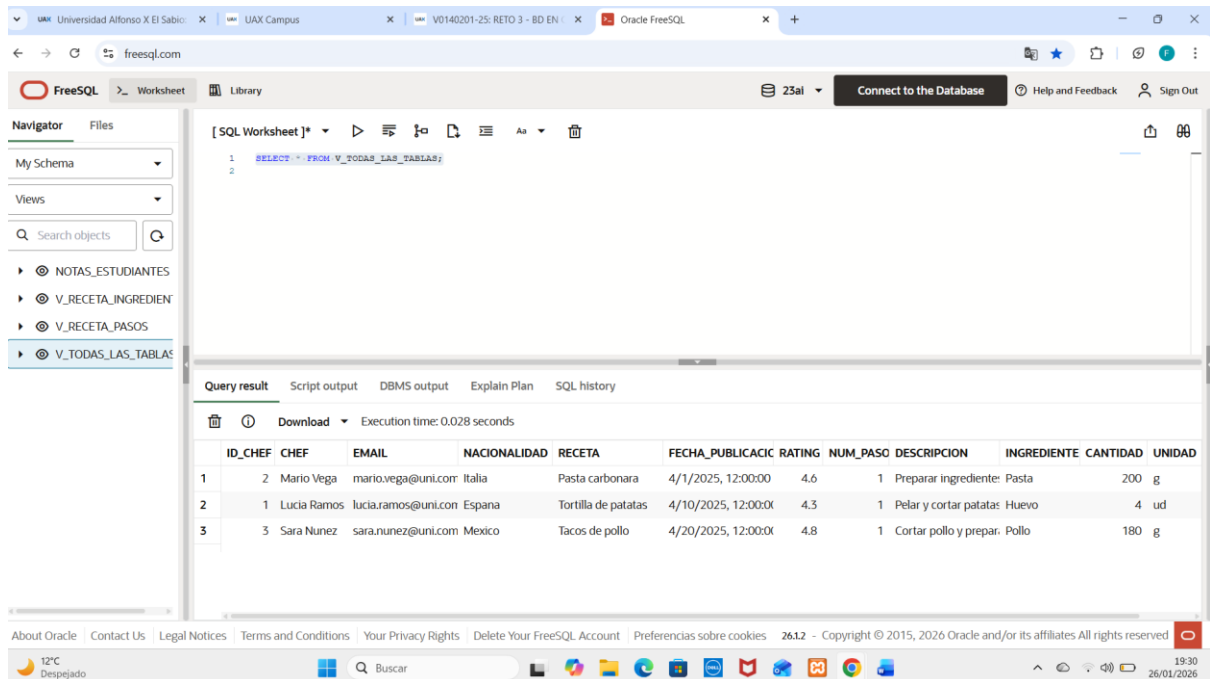
The screenshot shows the Oracle FreeSQL web interface with a SQL script in the main window. The script is as follows:

```
1 CREATE VIEW V_TODAS_LAS_TABLAS AS
2 SELECT C.ID AS ID_CHEF, C.NOMBRE AS CHEF, C.EMAIL, CP.NACIONALIDAD, R.TITULO AS RECETA,
3 R.FECHA_PUBLICACION, R.RATING, PA.NUM_PASO, PA.DESCRIPCION, I.NOMBRE AS INGREDIENTE,
4 RI.CANTIDAD, I.UNIDAD
5 FROM CHEF C INNER JOIN CHEF_PERFIL CP
6 ON C.ID = CP.ID_CHEF
7 INNER JOIN RECETA R
8 ON C.ID = R.ID_CHEF
9 INNER JOIN PASO PA
10 ON R.ID = PA.ID_RECETA
11 INNER JOIN RECETA_INGREDIENTE RI
12 ON R.ID = RI.ID_RECETA
13 INNER JOIN INGREDIENTE I
14 ON I.ID = RI.ID_INGREDIENTE;
```

The interface also shows the 'Script output' tab with the following message:

```
SQL> CREATE VIEW V_TODAS_LAS_TABLAS AS
SELECT C.ID AS ID_CHEF, C.NOMBRE AS CHEF, C.EMAIL, CP.NACIONALIDAD, R.TITULO AS RECETA,
R.FECHA_PUBLICACION, R.RATING, PA.NUM_PASO, PA.DESCRIPCION, I.NOMBRE AS INGREDIENTE,
RI.CANTIDAD, I.UNIDAD...
View V_TODAS_LAS_TABLAS created.
Elapsed: 00:00:00.013
```

The bottom of the page shows a footer with various links and a copyright notice for 2015, 2020 Oracle and/or its affiliates.



3.- Crear los siguientes procedimientos: (1 PUNTOS CADA UNA).

- PEGA AQUÍ LAS SENTENCIAS SQL -

ESTE PROCEDIMIENTO MOSTRARÁ POR CONSOLA UNA SUMA O MEDIA DE UNO DE LOS CAMPOS NUMÉRICOS

```
CREATE OR REPLACE PROCEDURE MEDIA_EXPERIENCIA_POR_NACIONALIDAD(P_NACIONALIDAD
IN VARCHAR2)
IS
    V_MEDIA NUMBER;
BEGIN
    SELECT AVG(EXPERIENCIA_ANIOS) INTO V_MEDIA FROM CHEF_PERFIL
    WHERE NACIONALIDAD = P_NACIONALIDAD;
    DBMS_OUTPUT.PUT_LINE('La media de experiencia para la nacionalidad ' || P_NACIONALIDAD
    || ' es: ' || V_MEDIA);
END;
```

- PEGA AQUÍ LAS CAPTURA DE PANTALLA COMPLETAS QUE MUESTREN SU CORRECTA EJECUCIÓN -

The screenshot shows the Oracle FreeSQL web interface. The left sidebar has a 'Navigator' tab with 'My Schema' selected. The main editor area contains the following SQL code:

```
1 CREATE OR REPLACE PROCEDURE MEDIA_EXPERIENCIA_POR_NACIONALIDAD(P_NACIONALIDAD IN VARCHAR2)
2 IS
3   V_MEDIA NUMBER;
4 BEGIN
5   SELECT AVG(EXPERIENCIA_ANIOS) INTO V_MEDIA FROM CHEF_PERFIL
6   WHERE NACIONALIDAD = P_NACIONALIDAD;
7 DBMS_OUTPUT.PUT_LINE(
8   'La media de experiencia para la nacionalidad ' || P_NACIONALIDAD || ' es: ' || V_MEDIA
9 );
10 END;
```

The 'Script output' tab is active, showing the execution results:

```
SQL> CREATE OR REPLACE PROCEDURE MEDIA_EXPERIENCIA_POR_NACIONALIDAD(P_NACIONALIDAD IN VARCHAR2)
IS
V_MEDIA NUMBER;
BEGIN...
Show more...

Procedure MEDIA_EXPERIENCIA_POR_NACIONALIDAD compiled
Elapsed: 00:00:00.155
```

The screenshot shows the Oracle FreeSQL web interface with the 'Procedures' tab selected in the left sidebar. The main editor area contains the following SQL code:

```
1 EXEC MEDIA_EXPERIENCIA_POR_NACIONALIDAD('Italia');
2
```

The 'Script output' tab is active, showing the execution results:

```
SQL> EXEC MEDIA_EXPERIENCIA_POR_NACIONALIDAD('Italia')

La media de experiencia para la nacionalidad Italia es: 6

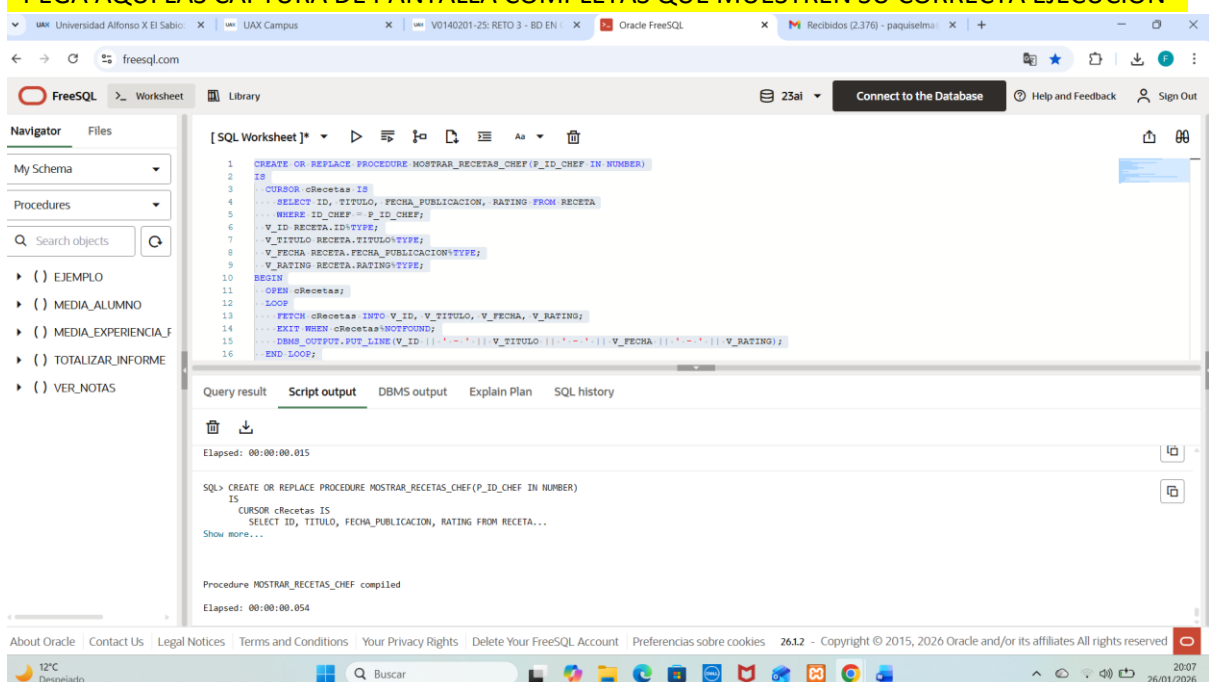
PL/SQL procedure successfully completed.
Elapsed: 00:00:00.015
```

ESTE PROCEDIMIENTO MOSTRARÁ TODOS LOS REGISTROS DE UNA TABLA QUE CUMPLAN CON UNA CONDICIÓN QUE SE ESTABLECERÁ POR PARÁMETRO. USA UN CURSOR EXPLÍCITO.

- PEGA AQUÍ LAS SENTENCIAS SQL -

```
CREATE OR REPLACE PROCEDURE MOSTRAR_RECETAS_CHEF(P_ID_CHEF IN NUMBER)
IS
  CURSOR cRecetas IS
    SELECT ID, TITULO, FECHA_PUBLICACION, RATING FROM RECETA
    WHERE ID_CHEF = P_ID_CHEF;
  V_ID RECETA.ID%TYPE;
  V_TITULO RECETA.TITULO%TYPE;
  V_FECHA RECETA.FECHA_PUBLICACION%TYPE;
  V_RATING RECETA.RATING%TYPE;
BEGIN
  OPEN cRecetas;
  LOOP
    FETCH cRecetas INTO V_ID, V_TITULO, V_FECHA, V_RATING;
    EXIT WHEN cRecetas%NOTFOUND;
    DBMS_OUTPUT.PUT_LINE(V_ID || ' - ' || V_TITULO || ' - ' || V_FECHA || ' - ' || V_RATING);
  END LOOP;
  CLOSE cRecetas;
EXCEPTION
  WHEN OTHERS THEN DBMS_OUTPUT.PUT_LINE('ERROR');
END;
```

- PEGA AQUÍ LAS CAPTURA DE PANTALLA COMPLETAS QUE MUESTREN SU CORRECTA EJECUCIÓN -



The screenshot displays the FreeSQL web application. The top navigation bar includes the FreeSQL logo and various utility links. The left sidebar shows a file explorer with a 'My Schema' dropdown and a search bar. The main area contains a SQL editor with the following code:

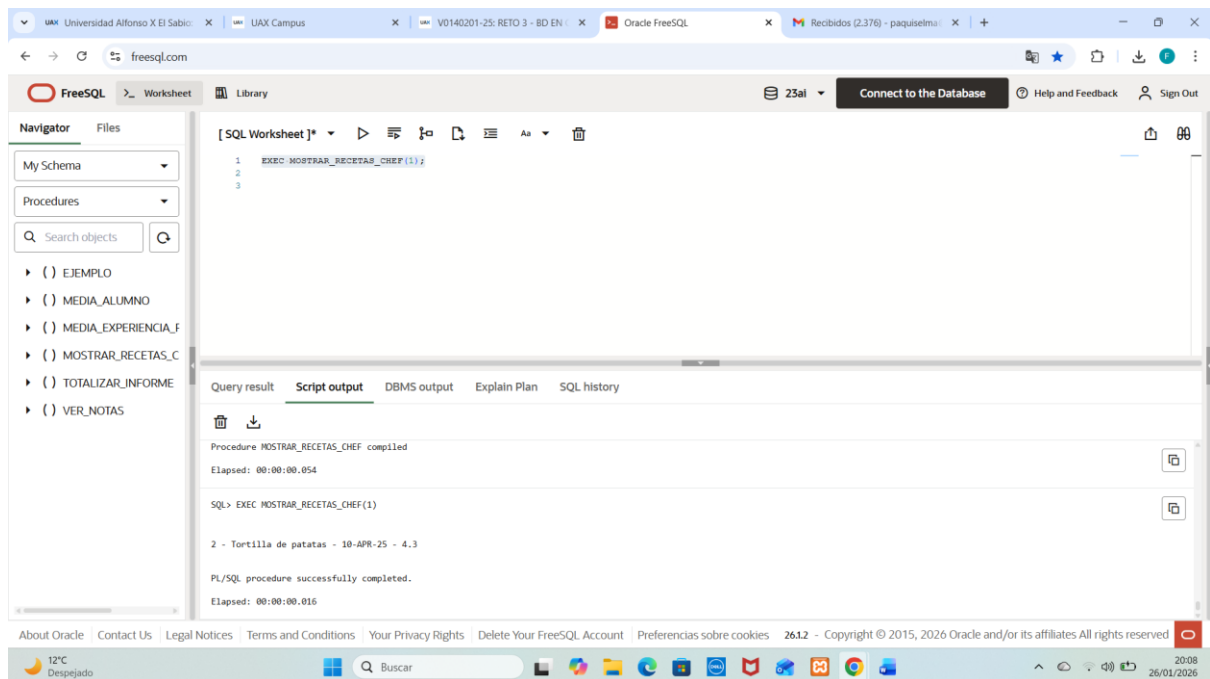
```
1 CREATE OR REPLACE PROCEDURE MOSTRAR_RECETAS_CHEF(P_ID_CHEF IN NUMBER)
2 IS
3   CURSOR cRecetas IS
4     SELECT ID, TITULO, FECHA_PUBLICACION, RATING FROM RECETA
5     WHERE ID_CHEF = P_ID_CHEF;
6   V_ID RECETA.ID%TYPE;
7   V_TITULO RECETA.TITULO%TYPE;
8   V_FECHA RECETA.FECHA_PUBLICACION%TYPE;
9   V_RATING RECETA.RATING%TYPE;
10 BEGIN
11   OPEN cRecetas;
12   LOOP
13     FETCH cRecetas INTO V_ID, V_TITULO, V_FECHA, V_RATING;
14     EXIT WHEN cRecetas%NOTFOUND;
15     DBMS_OUTPUT.PUT_LINE(V_ID || ' - ' || V_TITULO || ' - ' || V_FECHA || ' - ' || V_RATING);
16   END LOOP;
```

Below the editor, the 'Script output' tab is active, showing the execution results:

```
SQL> CREATE OR REPLACE PROCEDURE MOSTRAR_RECETAS_CHEF(P_ID_CHEF IN NUMBER)
IS
  CURSOR cRecetas IS
    SELECT ID, TITULO, FECHA_PUBLICACION, RATING FROM RECETA...
Show more...

Procedure MOSTRAR_RECETAS_CHEF compiled
Elapsed: 00:00:00.054
```

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4.- Crear un JOB que ejecute un PROCEDIMIENTO que inserte un registro en una tabla cada 10 minutos. (1 PUNTO)

- PEGA AQUÍ LAS SENTENCIAS SQL -

Se ha creado un procedimiento que inserta un registro en una tabla de control. Mediante un job programado con DBMS_SCHEDULER, dicho procedimiento se ejecuta automáticamente cada 10 minutos.

CREAMOS TABLA

```
CREATE TABLE CONTROL_JOB (  
  ID NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
  FECHA_INSERTION DATE  
);
```

CREAMOS PROCEDIMIENTO

```
CREATE OR REPLACE PROCEDURE INSERTAR_CONTROL_JOB IS  
BEGIN  
  INSERT INTO CONTROL_JOB (FECHA_INSERTION)  
  VALUES (SYSDATE);  
END;
```

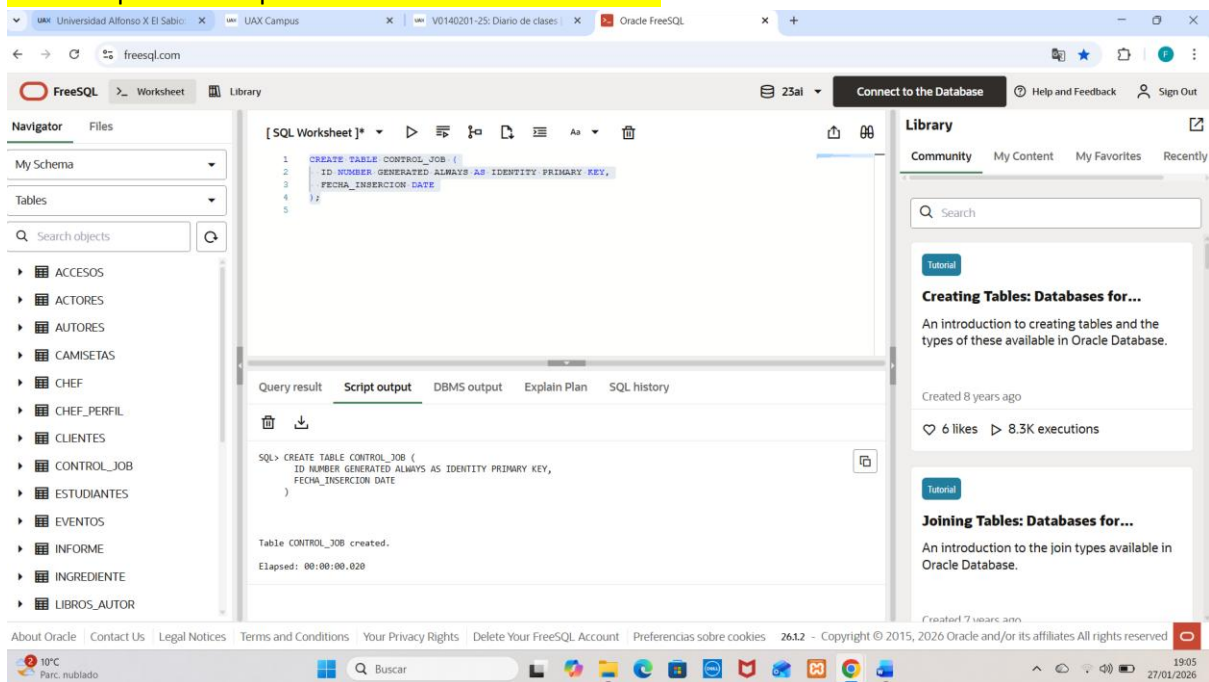
JOB EJECUCIÓN CADA 10 MINUTOS

```
BEGIN
DBMS_SCHEDULER.create_job (
  job_name      => 'job_cada_10_minutos',
  job_type      => 'STORED_PROCEDURE',
  job_action    => 'INSERTAR_CONTROL_JOB',
  start_date    => SYSTIMESTAMP,
  repeat_interval => 'FREQ=MINUTELY; INTERVAL=10',
  enabled       => TRUE
);
END;
```

```
SELECT * FROM CONTROL_JOB;
```

- PEGA AQUÍ LAS CAPTURA DE PANTALLA COMPLETAS QUE MUESTREN SU CORRECTA EJECUCIÓN -

No se ha podido comprobar con esta versión de Oracle



The screenshot shows the Oracle FreeSQL web interface. The top navigation bar includes the FreeSQL logo, a 'Worksheet' tab, and a 'Library' section. The left sidebar shows a 'Navigator' with 'My Schema' and a list of tables: ACCESOS, ACTORES, AUTORES, CAMISETAS, CHEF, CHEF_PERFIL, CLIENTES, CONTROL_JOB, ESTUDIANTES, EVENTOS, INFORME, INGREDIENTE, and LIBROS_AUTOR. The main area displays a SQL script in a 'Worksheet' editor. The script consists of two parts: a table creation and an insert procedure. The first part creates a table named 'CONTROL_JOB' with columns 'ID' (NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY) and 'FECHA_INSERCION' (DATE). The second part is a procedure named 'INSERTAR_CONTROL_JOB' that inserts a row into the 'CONTROL_JOB' table with the current date and time. The script is executed, and the 'Script output' tab shows the results: 'SQL> CREATE OR REPLACE PROCEDURE INSERTAR_CONTROL_JOB IS BEGIN INSERT INTO CONTROL_JOB (FECHA_INSERCION) VALUES (SYSDATE);... Show more...' and 'Procedure INSERTAR_CONTROL_JOB compiled'. The 'Elapsed: 00:00:00.020' and 'Elapsed: 00:00:00.054' are also displayed. The bottom status bar shows the Oracle logo, contact information, legal notices, and a copyright notice for 2015, 2026 Oracle and/or its affiliates.

The screenshot shows the Oracle FreeSQL web interface with an error message. The SQL script in the 'Worksheet' editor is the same as in the previous screenshot, but it includes an additional step: 'DBMS_SCHEDULER.create_job'. The script is executed, and the 'Script output' tab shows the error: 'ORA-27486: insufficient privileges ORA-06512: at 'SYS.DBMS_ISCHED', line 191 ORA-06512: at 'SYS.DBMS_SCHEDULER', line 332 ORA-06512: at line 2'. The error message is displayed in red text. The 'Elapsed: 00:00:00.020' and 'Elapsed: 00:00:00.054' are also displayed. The bottom status bar shows the Oracle logo, contact information, legal notices, and a copyright notice for 2015, 2026 Oracle and/or its affiliates.

5.- Crear mediante trigger un sistema que guarde un control de uso de una tabla. El sistema deberá reflejar que se ha insertado un registro, modificado o eliminado con sus datos y la fecha en la que se realiza la operación. No se admite multitrigger. Este enunciado se explicará en clase con un ejemplo. (2 PUNTOS)

- PEGA AQUÍ LAS SENTENCIAS SQL -

CREAMOS TABLA RECETA_CONTROL

```
CREATE TABLE RECETA_CONTROL(  
  ID NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
  OPERACION VARCHAR2(10),  
  FECHA DATE DEFAULT SYSDATE,  
  ID_RECETA NUMBER,  
  TITULO VARCHAR2(60),  
  FECHA_PUBLICACION DATE,  
  RATING FLOAT,  
  ID_CHEF NUMBER  
);
```

TRIGGER INSERT

```
CREATE OR REPLACE TRIGGER TRG_RECETA_INSERT  
AFTER INSERT ON RECETA  
FOR EACH ROW  
BEGIN  
  INSERT INTO RECETA_CONTROL(OPERACION, FECHA, ID_RECETA, TITULO,  
    FECHA_PUBLICACION, RATING, ID_CHEF)  
    VALUES ('INSERT', SYSDATE, :NEW.ID, :NEW.TITULO, :NEW.FECHA_PUBLICACION,  
    :NEW.RATING, :NEW.ID_CHEF);  
END;
```

```
INSERT INTO RECETA (TITULO, FECHA_PUBLICACION, RATING, ID_CHEF)  
VALUES ('Receta prueba trigger', SYSDATE, 4.2, 1);
```

```
SELECT * FROM RECETA_CONTROL;
```

TRIGGER UPDATE

```
CREATE OR REPLACE TRIGGER TRG_RECETA_UPDATE
AFTER UPDATE ON RECETA
FOR EACH ROW
BEGIN
    INSERT INTO RECETA_CONTROL(OPERACION, FECHA, ID_RECETA, TITULO,
    FECHA_PUBLICACION, RATING, ID_CHEF)
    VALUES
    ('UPDATE',SYSDATE,:NEW.ID,:NEW.TITULO,:NEW.FECHA_PUBLICACION,:NEW.RATING,:NEW.ID_
    CHEF);
END;
```

```
UPDATE RECETA
SET RATING = 4.8
WHERE TITULO = 'Receta prueba trigger';
```

```
SELECT * FROM RECETA_CONTROL;
```

TRIGGER DELETE

```
CREATE OR REPLACE TRIGGER TRG_RECETA_DELETE
BEFORE DELETE ON RECETA
FOR EACH ROW
BEGIN

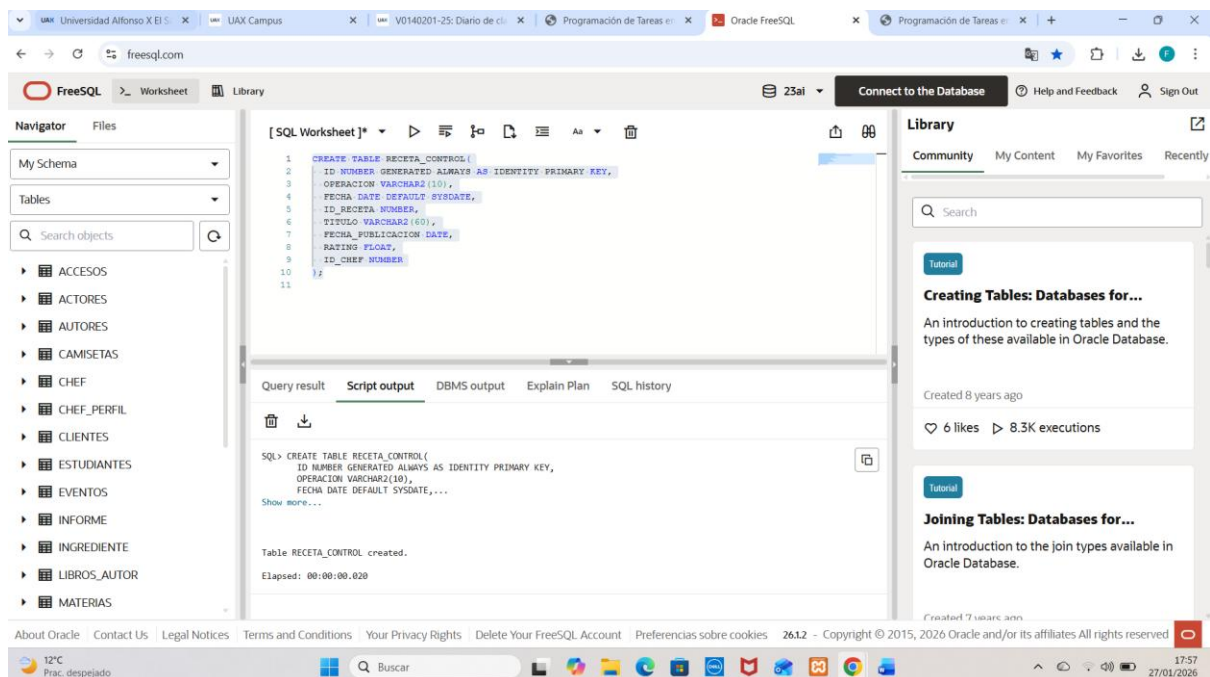
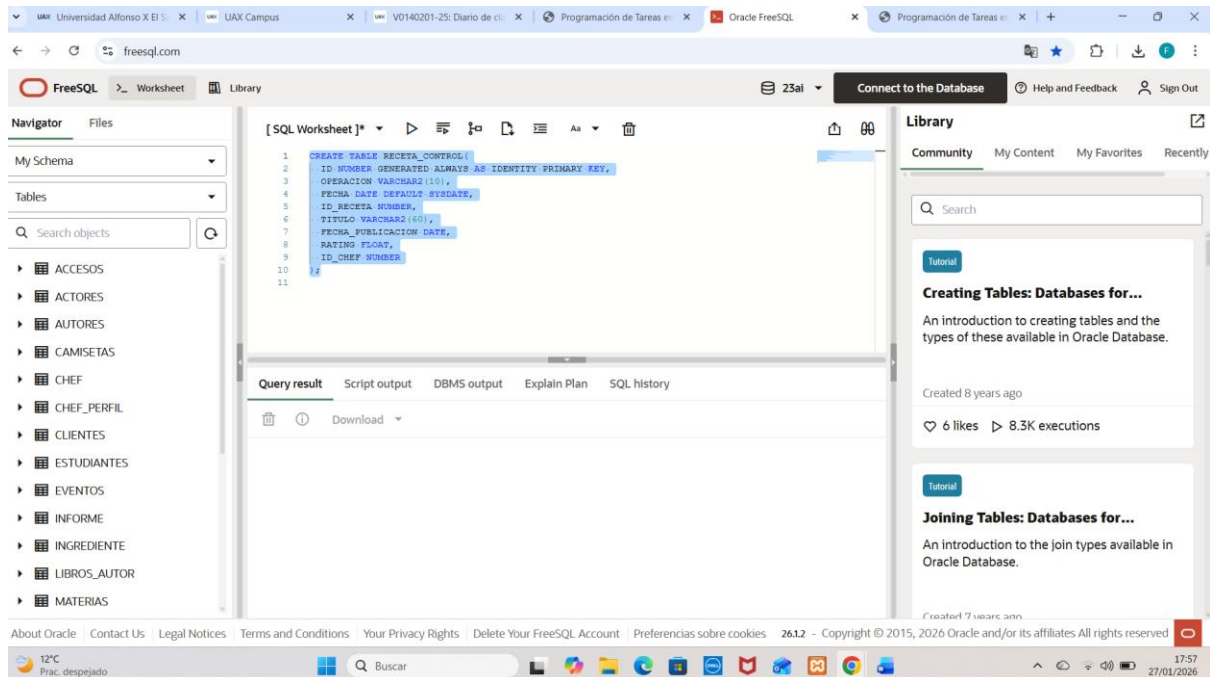
    INSERT INTO RECETA_CONTROL(OPERACION, FECHA, ID_RECETA, TITULO,
    FECHA_PUBLICACION, RATING, ID_CHEF)
    VALUES('DELETE', SYSDATE,:OLD.ID,:OLD.TITULO,:OLD.FECHA_PUBLICACION,:OLD.RATING,
    :OLD.ID_CHEF);
END;
```

```
DELETE FROM RECETA
WHERE TITULO = 'Receta prueba trigger';
```

```
SELECT * FROM RECETA_CONTROL;
```

- PEGA AQUÍ LAS CAPTURA DE PANTALLA COMPLETAS QUE MUESTREN SU CORRECTA EJECUCIÓN -

CREAMOS LA TABLA RECETA_CONTROL



TRG_RECETA_INSERT

The screenshot shows the Oracle FreeSQL web interface. The left sidebar contains a 'Navigator' with 'My Schema' and 'Triggers' sections. The main editor area displays the SQL script for creating the trigger:

```
1 CREATE OR REPLACE TRIGGER TRG_RECETA_INSERT
2 AFTER INSERT ON RECETA
3 FOR EACH ROW
4 BEGIN
5     INSERT INTO RECETA_CONTROL (OPERACION, FECHA, ID_RECETA, TITULO, FECHA_PUBLICACION, RATING, ID_CHEF)
6     VALUES ('INSERT', SYSDATE, :NEW.ID, :NEW.TITULO, :NEW.FECHA_PUBLICACION, :NEW.RATING, :NEW.ID_CHEF);
7 END;
```

The 'Script output' tab shows the execution results:

```
SQL> CREATE OR REPLACE TRIGGER TRG_RECETA_INSERT
  AFTER INSERT ON RECETA
  FOR EACH ROW
  BEGIN ...
Trigger TRG_RECETA_INSERT compiled
Elapsed: 00:00:00.057
```

The screenshot shows the Oracle FreeSQL web interface. The left sidebar contains a 'Navigator' with 'My Schema' and 'Triggers' sections. The main editor area displays the SQL script for inserting data into the RECETA table:

```
1 INSERT INTO RECETA (TITULO, FECHA_PUBLICACION, RATING, ID_CHEF)
2 VALUES ('Receta prueba trigger', SYSDATE, 4.2, 1);
3
4
5
6
```

The 'Script output' tab shows the execution results:

```
Trigger TRG_RECETA_DELETE compiled
Elapsed: 00:00:00.055

SQL> INSERT INTO RECETA (TITULO, FECHA_PUBLICACION, RATING, ID_CHEF)
  VALUES ('Receta prueba trigger', SYSDATE, 4.2, 1)

1 row inserted.
Elapsed: 00:00:00.013
```

The screenshot shows the Oracle FreeSQL web interface. The top navigation bar includes links for 'My Schema', 'Triggers', and a search bar. The main workspace displays a query result table with the following data:

ID	OPERACION	FECHA	ID_RECETA	TITULO	FECHA_PUBLICACION	RATING	ID_CHEF
1	21 INSERT	1/28/2026, 5:53:57	41	Receta prueba trigge	1/28/2026, 5:53:57	4.2	1
2	1 INSERT	1/27/2026, 5:01:48	21	Receta prueba trigge	1/27/2026, 5:01:48	3.5	1

TRG_RECETA_UPDATE

The screenshot shows the Oracle FreeSQL web interface with the SQL script for creating a trigger. The script is as follows:

```
1 CREATE OR REPLACE TRIGGER TRG_RECETA_UPDATE
2 AFTER UPDATE ON RECETA
3 FOR EACH ROW
4 BEGIN
5     INSERT INTO RECETA_CONTROL(OPERACION, FECHA, ID_RECETA, TITULO, FECHA_PUBLICACION, RATING, ID_CHEF)
6     VALUES ('UPDATE',SYSDATE,:NEW.ID,:NEW.TITULO,:NEW.FECHA_PUBLICACION,:NEW.RATING,:NEW.ID_CHEF);
7 END;
```

The 'Script output' tab shows the execution details:

```
SQL> CREATE OR REPLACE TRIGGER TRG_RECETA_UPDATE
AFTER UPDATE ON RECETA
FOR EACH ROW
BEGIN ...
Trigger TRG_RECETA_UPDATE compiled
Elapsed: 00:00:00.055
```

The screenshot shows the Oracle FreeSQL web interface. The SQL Worksheet contains the following query:

```
1 UPDATE RECETA
2 SET RATING = 4.8
3 WHERE TITULO = 'Receta prueba trigger';
4
5
6
7
8
9
```

The Query result tab shows the following output:

```
0 rows updated.
Elapsed: 00:00:00.002

SQL> UPDATE RECETA
SET RATING = 4.8
WHERE TITULO = 'Receta prueba trigger'

2 rows updated.
Elapsed: 00:00:00.009
```

The screenshot shows the Oracle FreeSQL web interface. The SQL Worksheet contains the following query:

```
1 SELECT * FROM RECETA_CONTROL;
2
3
4
5
6
7
8
```

The Query result tab shows the following output:

```
Download Execution time: 0.001 seconds
```

ID	OPERACION	FECHA	ID_RECETA	TITULO	FECHA_PUBLICACI	RATING	ID_CHEF
1	21 INSERT	1/28/2026, 5:53:57	41	Receta prueba trigge	1/28/2026, 5:53:57	4.2	1
2	41 UPDATE	1/28/2026, 6:03:45	41	Receta prueba trigge	1/28/2026, 5:53:57	4.8	1
3	42 UPDATE	1/28/2026, 6:03:45	21	Receta prueba trigge	1/27/2026, 5:01:48	4.8	1
4	1 INSERT	1/27/2026, 5:01:48	21	Receta prueba trigge	1/27/2026, 5:01:48	3.5	1

TRG_RECETE_DELETE

The screenshot shows the Oracle FreeSQL web interface. The left sidebar contains a 'Navigator' panel with 'My Schema' selected, showing a list of triggers: COPIAR, RESTAR_LIBRO, SUMAR_LIBRO, TRG_RECETA_INSERT, and TRG_RECETA_DELETE. The main editor area displays the SQL code for creating the trigger:

```
1 CREATE OR REPLACE TRIGGER TRG_RECETA_DELETE
2 BEFORE DELETE ON RECETA
3 FOR EACH ROW
4 BEGIN
5
6   INSERT INTO RECETA_CONTROL (OPERACION, FECHA, ID_RECETA, TITULO, FECHA_PUBLICACION, RATING, ID_CHEF)
7   VALUES ('DELETE', SYSDATE, :OLD.ID, :OLD.TITULO, :OLD.FECHA_PUBLICACION, :OLD.RATING, :OLD.ID_CHEF);
8 END;
```

The 'Script output' tab is active, showing the execution results:

```
SQL> CREATE OR REPLACE TRIGGER TRG_RECETA_DELETE
BEFORE DELETE ON RECETA
FOR EACH ROW
BEGIN ...
Show more...

Trigger TRG_RECETA_DELETE compiled
Elapsed: 00:00:00.055
```

The screenshot shows the Oracle FreeSQL web interface after the trigger has been created. The left sidebar now includes TRG_RECETA_DELETE and TRG_RECETA_UPDATE. The main editor area displays the SQL code for deleting a recipe:

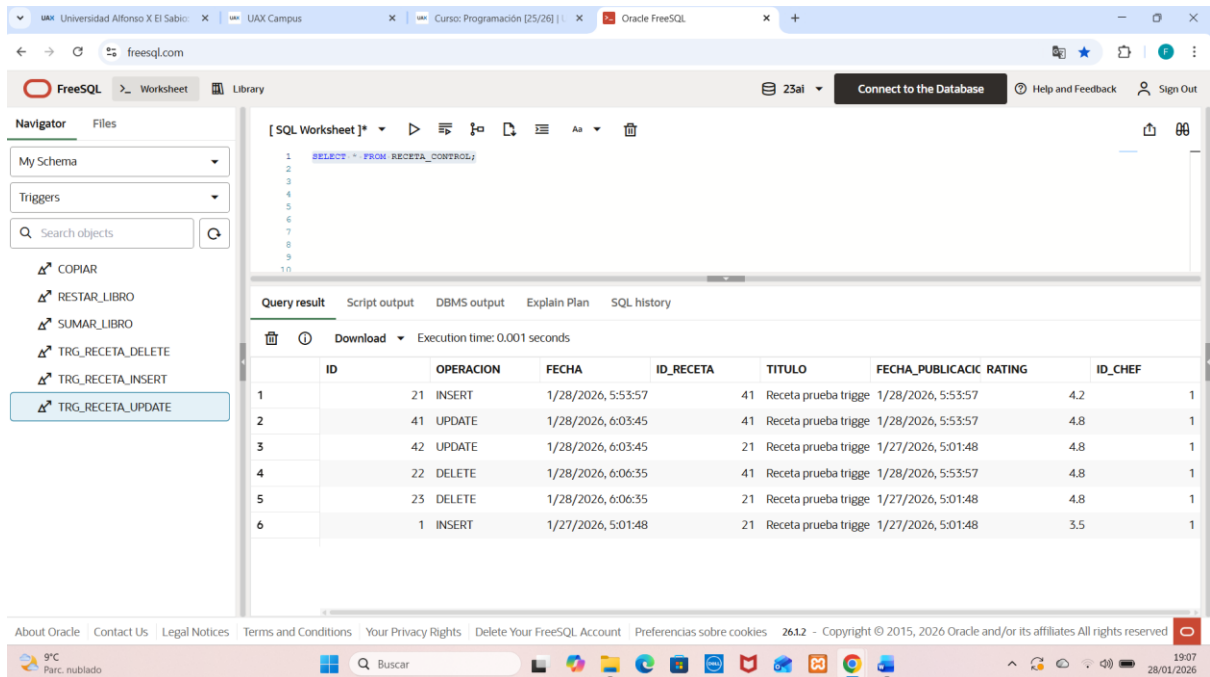
```
1 DELETE FROM RECETA
2 WHERE TITULO = 'Receta prueba trigger';
```

The 'Script output' tab is active, showing the execution results:

```
2 rows updated.
Elapsed: 00:00:00.009

SQL> DELETE FROM RECETA
WHERE TITULO = 'Receta prueba trigger'

2 rows deleted.
Elapsed: 00:00:00.014
```



The screenshot shows the Oracle FreeSQL web interface. The top navigation bar includes the Oracle logo, 'FreeSQL', and tabs for 'Worksheet' and 'Library'. The left sidebar contains a 'Navigator' section with 'My Schema' and 'Triggers' dropdowns, a search bar, and a list of actions: 'COPIAR', 'RESTAR_LIBRO', 'SUMAR_LIBRO', 'TRG_RECETA_DELETE', 'TRG_RECETA_INSERT', and 'TRG_RECETA_UPDATE'. The main area displays a query result table with columns: ID, OPERACION, FECHA, ID_RECETA, TITULO, FECHA_PUBLICACION, RATING, and ID_CHEF. The table contains 6 rows of data. The bottom of the interface shows a footer with links like 'About Oracle', 'Contact Us', and 'Legal Notices', along with a copyright notice for 2015 and 2026. The Windows taskbar at the bottom shows the date as 28/01/2026 and the time as 19:07.

ID	OPERACION	FECHA	ID_RECETA	TITULO	FECHA_PUBLICACION	RATING	ID_CHEF
1	21 INSERT	1/28/2026, 5:53:57	41	Receta prueba trigge	1/28/2026, 5:53:57	4.2	1
2	41 UPDATE	1/28/2026, 6:03:45	41	Receta prueba trigge	1/28/2026, 5:53:57	4.8	1
3	42 UPDATE	1/28/2026, 6:03:45	21	Receta prueba trigge	1/27/2026, 5:01:48	4.8	1
4	22 DELETE	1/28/2026, 6:06:35	41	Receta prueba trigge	1/28/2026, 5:53:57	4.8	1
5	23 DELETE	1/28/2026, 6:06:35	21	Receta prueba trigge	1/27/2026, 5:01:48	4.8	1
6	1 INSERT	1/27/2026, 5:01:48	21	Receta prueba trigge	1/27/2026, 5:01:48	3.5	1

IMPORTANTE:

Se entrega un único documento en pdf, no se revisarán trabajos en otro formato.

Si la captura de pantalla no se adjunta o no es de pantalla completa con fecha y hora a la vista se anulará el 100% del valor de la pregunta.

No se admite uso de IA ni sintaxis diferente a la vista en clase, en caso contrario se calificará el ejercicio con un cero.